

✓

Pitchfork

bifurcation

$$\dot{x} = x(r - x^2)$$

$$\dot{x} = f(x, r)$$

$$\frac{\partial f(x, r)}{\partial x} \bigg|_{x=x^*}$$

$> 0$ , unstable

$< 0$ , stable

$$\begin{aligned} \frac{\partial f}{\partial x} &= \frac{\partial}{\partial x} (xr - x^3) = \\ &= r - 3x^2 \end{aligned}$$

$$x^* = 0,$$

$$\frac{\partial f}{\partial x} \bigg|_{x=0} = r$$

$> 0$  inst

$< 0$  stable

2'

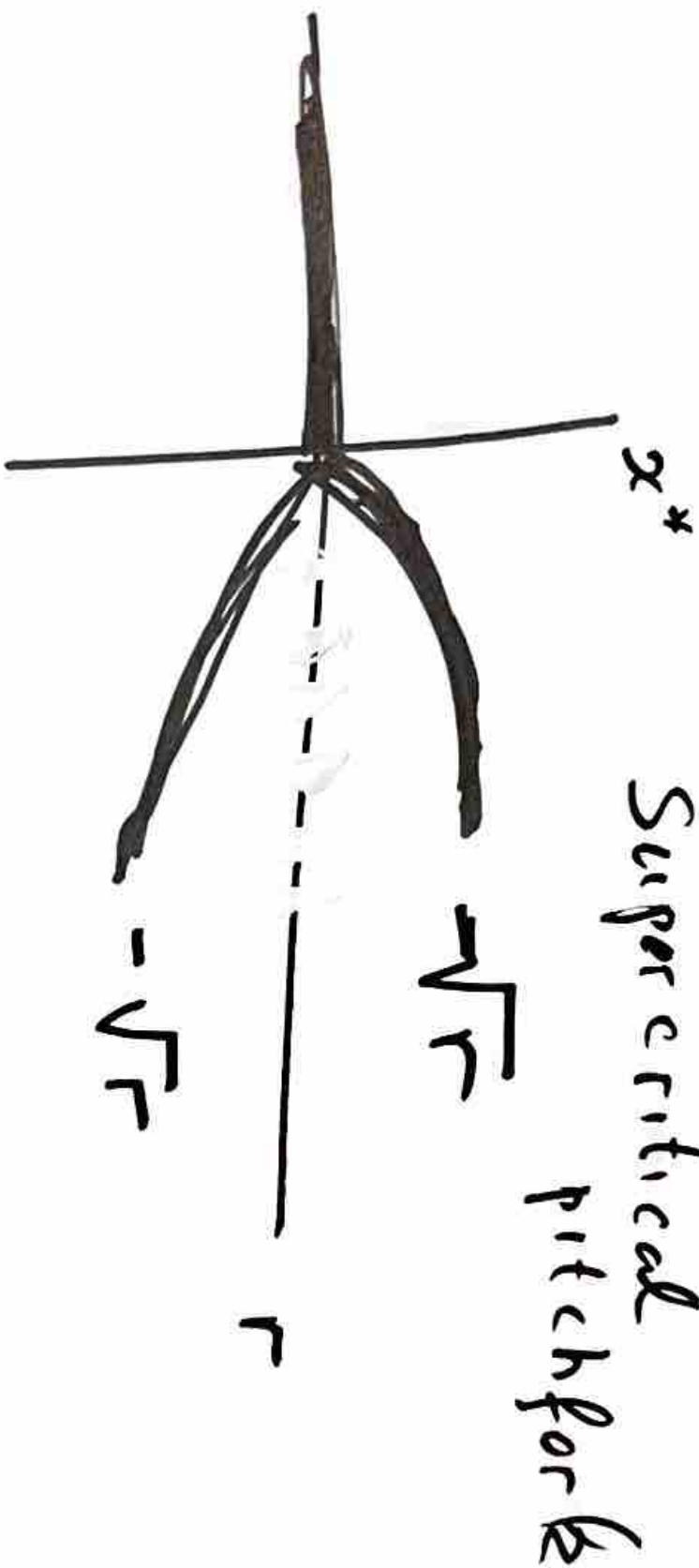
$$x_{2,3}^* = \pm \sqrt{r}, \text{ exists } r > 0,$$

$$\frac{\partial f}{\partial x} \Big|_{x = \pm \sqrt{r}} = r - 3r = -2r$$

$r > 0$ , stable

$r < 0$  unstable X

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Analyze stability,  
plot bifurcation  
diagram

$$\dot{x} = -x + \beta \tanh x$$

$$\dot{x} = -x + \beta \tanh(x)$$

Solution

$$x = \beta \tanh(x)$$

$$\beta = \frac{x}{\tanh x}$$

$$x^* = 0: \quad \dot{x} = 0 + \beta \tanh(0) = 0$$

If  $|x| \ll 1$ ,

$$\tanh(x) \approx x - \frac{x^3}{3}$$

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$$\text{If } |x| \ll 1, \tanh x \approx x - \frac{x^3}{3}$$

$$\dot{x} = -x + \beta \tanh x$$

$$\approx -x + \beta \left( x - \frac{x^3}{3} \right)$$

$$= x(\beta - 1) - \beta \frac{x^3}{3}$$

$$\dot{x} = f(x, \beta)$$

$$f(x, \beta) = -x + \beta \tanh(x)$$

$$\frac{\partial f(x, \beta)}{\partial x} = -1 + \frac{\beta}{\cosh^2 x}$$

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$$x_1^* = 0,$$

$$\frac{\partial f}{\partial x} \Big|_{x^*=0} = -1 + \beta, \quad \beta > 1, \quad \text{unstable}$$

$$\dot{x} = x \left[ \beta - 1 - \beta \frac{x^2}{3} \right] \quad \beta < 1, \quad \text{stable}$$

$$[\ ] = 0 \Rightarrow \beta - 1 = \beta \cdot \frac{x^2}{3}$$

$$\pm \sqrt{3 \frac{\beta - 1}{\beta}} = x_{2,3}^* \quad |x| < 1$$

Exist. only if  $\beta > 1$

$$\frac{\partial f}{\partial x} = \beta - 1 - \beta x^2$$

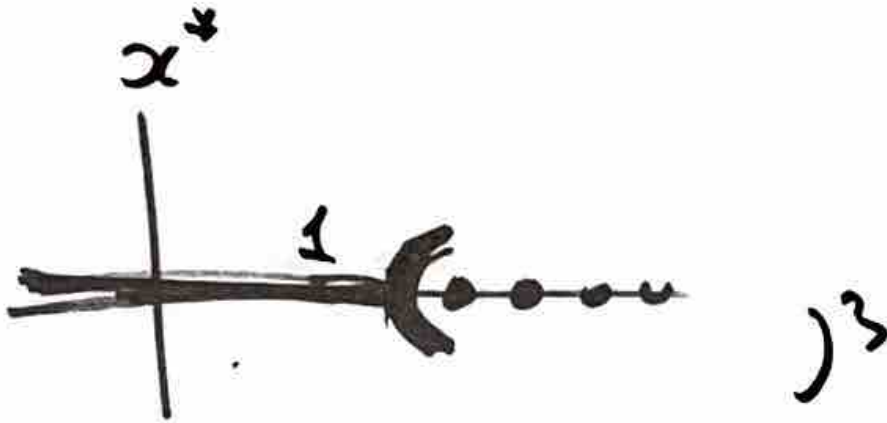
$$\frac{\partial f}{\partial x} \Big|_{x = \pm \sqrt{3 \frac{\beta - 1}{\beta}}} = \beta - 1 - 3(\beta - 1) =$$

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$$\mu > 1$$

$$\left. \frac{\partial f}{\partial x} \right|_{x=\pm\sqrt{\mu}} = -2(\mu-1)$$

$$= \text{stable} = -2(\mu-1)$$



$$\mu = \frac{x}{\tanh x}$$

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$$x_1^* = 0: \quad \left. \frac{\partial f}{\partial x} \right|_{x=0} = -1 + \beta$$

$\beta > 1$       unstable

$\beta < 1$       stable

$$x_{2,3}^* = \pm \sqrt{3 \left( \frac{\beta - 1}{\beta} \right)}$$

$$\left. \frac{\partial f}{\partial x} \right|_{x = \pm \sqrt{3 \left( \frac{\beta - 1}{\beta} \right)}} = -2(\beta - 1)$$

$\beta > 1$ , stable

$\beta < 1$ , unstable

"modulus Aaron"



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Example

Consider  $\dot{x} = x(r - x^2)$

Plot the potential for

$r > 0$ ,  $r = 0$ ,  $r < 0$

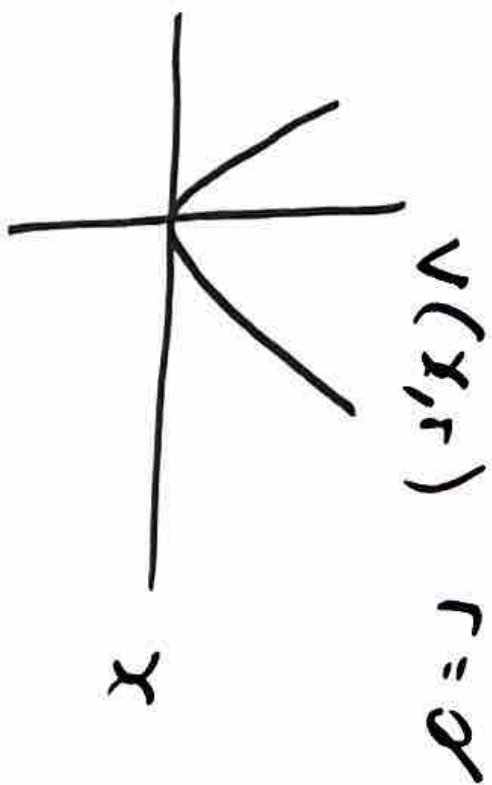
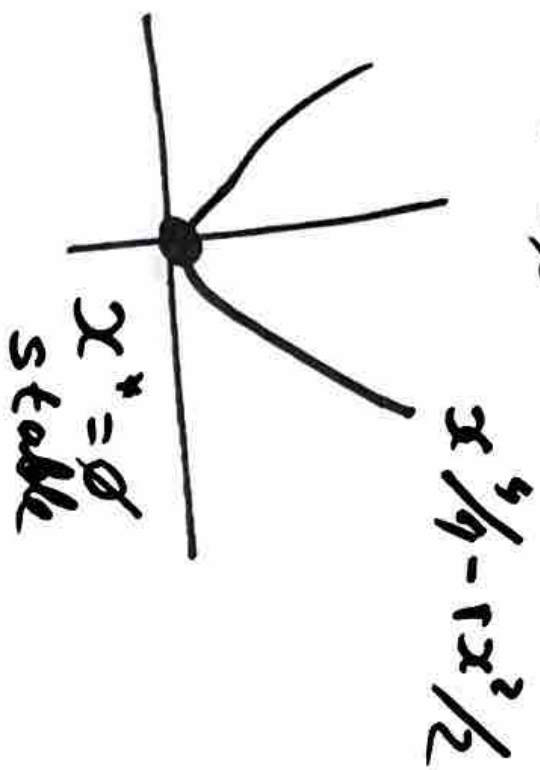
Solution

$$\dot{x} = xr - x^3 = -\frac{\partial V}{\partial x}$$

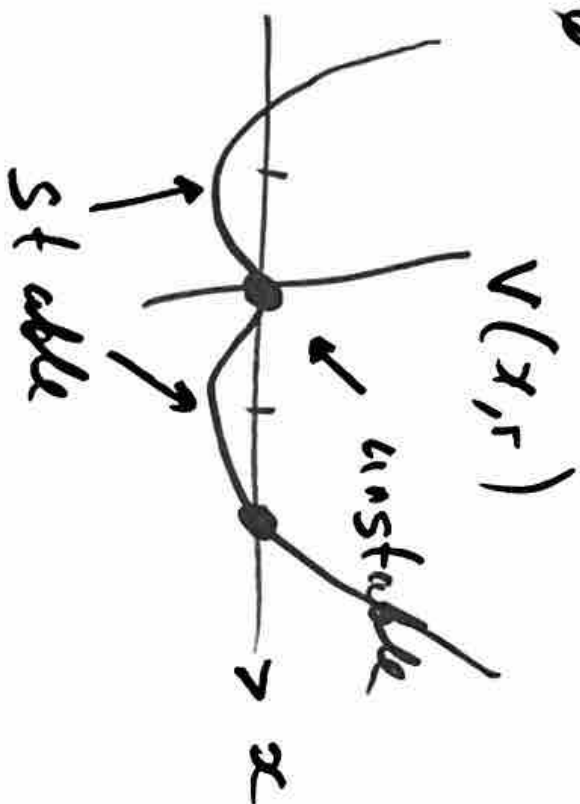
$$V(x, r) = -\frac{rx^2}{2} + \frac{x^4}{4} =$$

$$= \frac{x^2}{2} \left( \frac{x^2}{2} - r \right)$$

$$r < 0$$



$$r > 0$$



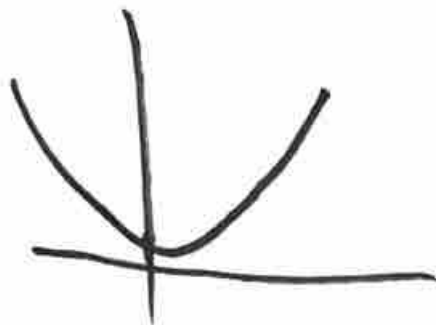
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$$\dot{x} = x(r - x^2) = xr - x^3$$

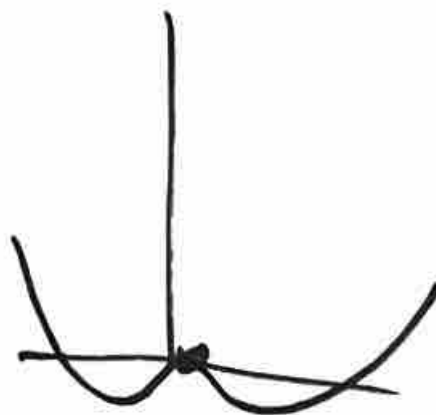
$\uparrow$                        $\uparrow$   
                                  minus

$$r \leq 0$$

$$V(x)$$



$$r > 0$$



-12-  $\dot{x} = x(r - x^2)$

Subcritical Pitchfork

$$\dot{x} = x(r + x^2)$$

$$x_1^* = 0$$

$$x_{2,3}^* = \pm \sqrt{-r}, \quad \text{Aaron: } r < 0$$

$$\dot{x} = f(x, r) = x(r + x^2)$$

$$\frac{\partial f(x, r)}{\partial x} = r + 3x^2$$

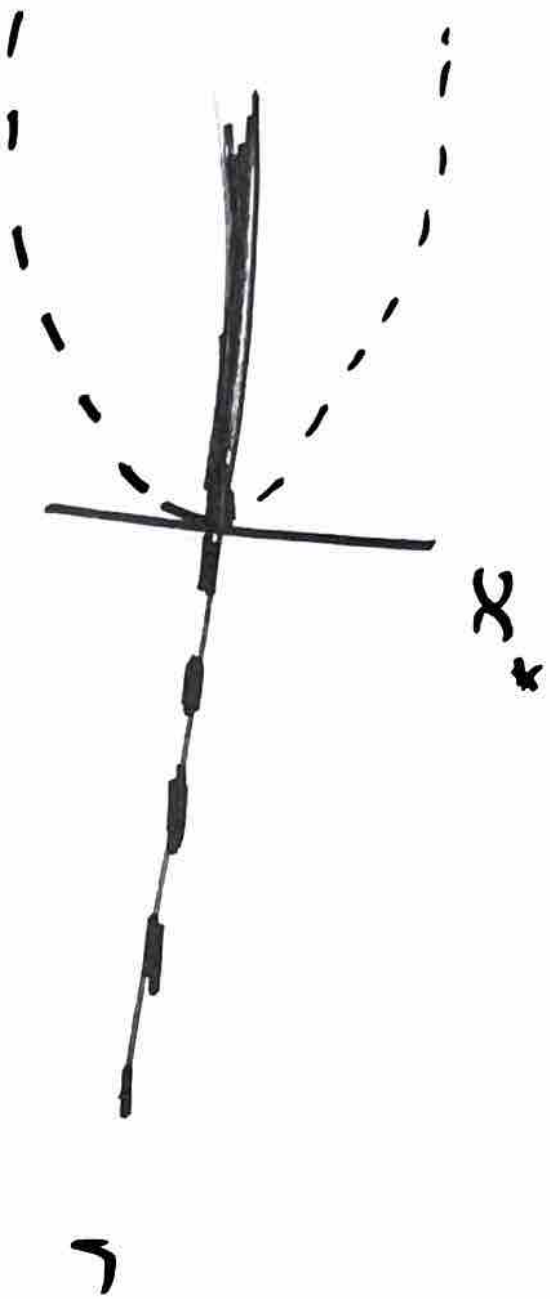
$$x_1^* = 0, \quad \frac{\partial f}{\partial x} = r$$

- stable,  $r < 0$   
unstable,  $r > 0$

$$x_{2,3} = \pm \sqrt{-r}$$

$$\frac{\partial f}{\partial x} = r + 3(-r) = -2r, \quad r < 0 \text{ unstable}$$

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No stable  
fixed points  
no zeros

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Obtain ANALYTICAL  
Solution for  $\dot{x} = rx + x^3$   
Solution

$$\int \frac{dx}{rx + x^3} = \int dt = t + C$$

$$\frac{1}{rx + x^3} = \frac{A}{x} + \frac{B + Cx}{r + x^2}$$

$$= \frac{Ar + Ax^2 + Bx + Cx^2}{x(r + x^2)}$$

$$= \frac{x^2(A + C) + Bx + Ar}{x(r + x^2)}$$

$$A = \frac{1}{r} \quad B = 0 \quad C = -A = -\frac{1}{r}$$

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$$\frac{1}{rx+x^3} = \dots$$
$$= + \frac{1}{rx} - \frac{1}{r} \frac{x}{r+x^2}$$

$$\int \frac{dx}{rx+x^3} = \dots$$

$$= \int \frac{dx}{rx} - \int \frac{dx \cdot x}{r(r+x^2)}$$

$$= \frac{1}{r} \ln |x| - \frac{1}{2r} \ln |r+x^2|$$

$$= \frac{1}{r} \ln \left| \frac{x}{\sqrt{r+x^2}} \right| = \dots$$

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$$\ln \left| \frac{x}{\sqrt{r+x^2}} \right| = rt + rc$$

$$x = e^{rt+rc} \sqrt{r+x^2}$$

$$x^2 = e^{2rt+2rc} (r+x^2)$$

$$x^2 \left( \frac{1 - e^{2rt+2rc}}{r e^{2rt+2rc}} \right) = r e^{2rt+2rc}$$
$$x = \pm \sqrt{\frac{r e^{2rt+2rc}}{1 - e^{2rt+2rc}}}$$

$$x = \pm \sqrt{r} \cdot \frac{1}{\sqrt{e^{-2rt-2rc} - 1}}$$



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$$x(t=0) = \sqrt{r} \frac{1}{\sqrt{e^{-2rc} - 1}} = x_0$$

$$\left( \frac{\sqrt{r}}{x_0} \right)^2 = e^{-2rc} - 1$$

$$e^{-2rc} = 1 + \left( \frac{\sqrt{r}}{x_0} \right)^2$$

$$x = \pm \sqrt{r}$$

$$\sqrt{\left( 1 + \frac{r}{x_0^2} \right) e^{-2rt} - 1}$$

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Suppose  $\dot{x} = 1x - x^3$

$$\dot{x} = 1x + x^3 - x^5$$

-18-      Sol. per       $\dot{x} = 1x - x^3$

$$\dot{x} = 1x + x^3 - x^5$$

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$$\dot{x} = rx + x^3 - x^5$$

$$\dot{u} = au + bu^3 - cu^5$$

$$a, b, c > 0$$

$$u = \alpha x,$$

$$t = \beta \tau; \quad \frac{\partial}{\partial t} = \frac{\partial \tau}{\partial t} \frac{\partial}{\partial \tau} = \frac{1}{\beta} \frac{\partial}{\partial \tau}$$

$$\frac{d}{d\tau} \frac{\partial}{\partial \tau} x = a \alpha x + b \alpha^3 x^3 - c \alpha^5 x^5$$

$$\frac{\partial}{\partial \tau} x = a \beta x + \underbrace{b \beta \alpha^2}_{=1} x^3 - \underbrace{c \beta \alpha^4}_{=1} x^5$$

$$b \beta \alpha^2 = 1$$

$$c \beta \alpha^4 = 1$$

$$\frac{c \alpha^2}{b} = 1, \quad \alpha = \sqrt{b/c}$$

$$\beta = \frac{1}{\alpha^2 b} = \frac{c}{b^2}$$

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$$\dot{x} = rx - x^3$$



$$\dot{x} = rx + x^3$$



$$\dot{x} = rx + x^3 - x^5$$